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Enhanced Quadrotor Drone Control: Integration of Backstepping Method with Butterfly Optimization Algorithm

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***Abstract*—This paper presents a robust approach for controlling a quadrotor drone, using the backstepping control method combined with the Butterfly Optimization Algorithm (BOA) to optimize its parameters, thereby improving its stability, and maneuverability. Barrier Lyapunov Functions are employed to provide a rigorous stability analysis and guarantee safe operation within defined constraints. The study begins with the identification of the mathematical model that governs the dynamics of the system. Subsequently, a backstepping controller is employed to control the drone's behavior. To further enhance the controller's performance, the BOA technique is employed to optimize the parameters associated with the controller. To estimate the hidden states of the quadrotor system, a trianguler observer is used. In addition, Barrier Lyapunov is employed to verify the efficiency of the suggested approach in maintaining safety and stability. The simulation results demonstrate the efficacy of the suggested strategy in comparison with traditional control techniques.**

***Index Terms*—Backstepping control, barrier Lyapunov, butterfly optimization algorithm, drone quadrotor, triangular observer.**

I. Introduction

The quadrotor drone has become a significant and versatile technology, with applications in diverse fields including aerial surveillance, search and rescue missions, and agriculture [1], [2]. The key advantages of this technology are its agility, and ability to maneuver in complex environments [3]–[5]. Although quadrotor drones offer a multitude of advantages, they present significant control challenges due to their inherent nonlinear and multi-input multi-output (MIMO) dynamics. These systems are highly susceptible to external disturbances, and are particularly liable to instability in the absence of careful control [6], [7], [9], [10]. It is therefore imperative that robust and precise control strategies are employed to guarantee the stability, safety and performance of quadrotors in real-world applications [11], [12], [15].

The existing literature proposes a number of control methodologies with the objective of enhancing the stability and performance of quadrotor systems. The application of traditional control techniques, such as proportional-integral-derivative (PID) control, offering a degree of simplicity and ease of implementation. However, these methods frequently prove inadequate when confronted with the highly nonlinear and dynamic characteristics of quadrotors [16]. To overcome these limitations, more advanced strategies such as Sliding Mode Control (SMC) [8], [17]–[19] and fuzzy logic technique [20]–[23] have gained importance in the field of control theory due to their ability to handle nonlinearities and uncertainties more effectively. In particular, backstepping control has gained considerable attention for quadrotor systems, this approach involves dividing the system dynamics into manageable subsystems, thereby ensuring stability at each stage [24], [25]. The combination of backstepping with the Lyapunov approach has traditionally been employed to guarantee system stability and convergence to the desired trajectory [26], [27]. However, recent developments have introduced alternative methodologies, such as the deployment of barrier Lyapunov functions, which impose constraints on the system states to ensure they remain within predefined limits [28], [29]. Furthermore, it is notable that optimization algorithms have been extensively applied in order to dynamically adjust controller parameters in response to varying conditions and uncertainties. Among these, the Particle Swarm Optimization (PSO) approach and the Butterfly Optimization Algorithm (BOA), have demonstrated considerable potential for enhancing the efficiency and

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responsiveness of control systems [30], [31].
This paper advances the field of quadrotor drone control by integrating Barrier Lyapunov functions into the backstepping control system, ensuring rigorous stability and safety within predefined constraints [24]. The study employs the BOA to optimize the parameters of the backstepping controller, thereby considerably improving the drone's performance in terms of stability, robustness, and maneuverability [31]. It is acknowledged that not all states can be directly measured; thus, a triangular observer is introduced to the control system in order to estimate the hidden states [32]. The results of simulations demonstrate that the proposed approach outperforms traditional control methods, offering superior control performance and adaptability [33], [34].
The structure of this paper is as follows: The quadrotor drone's mathematical model is presented in Section 2. The design of the proposed control system is the primary focus of Section 3. The suggested triangular observer is described in section 4. The simulation results are shown in Section 5. A conclusion summarizing the results and suggesting future directions is provided in Section 6.

II. The Mathematical Model of the Drone Quadrotor

A particular focus is placed on adequately modeling the multi-input multi-output (MIMO) nonlinear dynamics of the quadrotor drone when developing its mathematical model. The motion of the drone is described in terms of its translational and rotational movements in three-dimensional space, which are governed by Newton-Euler equations [35], [36]. These equations of motion are divided into two principal components: the first concerns translational dynamics, which control the position of the drone in space, and the second pertains to rotational dynamics, which determine the orientation (roll, pitch, and yaw) of the drone [37], [38]. The model incorporates the effects of gravity, thrust from the four rotors, and gyroscopic forces that arise from the angular motion of the rotors [33].

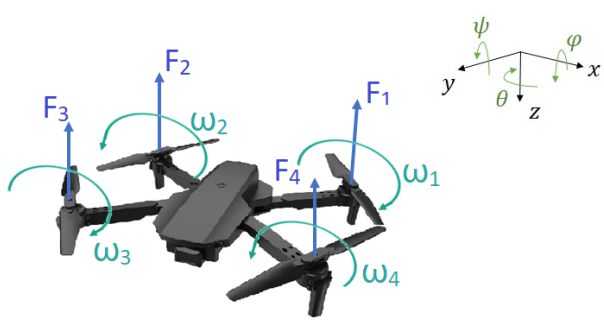

Fig. 1: Model of the quadrotor drone

The mathematical representation of the quadrotor system is expressed using Newton-Euler formalism as a set of non-linear differential equations that describe both the translational and rotational dynamics by a state vector $\underline{w} = [x\ y\ z\ \phi\ \theta\ \psi]$, in the following form:

$$\ddot{w}_i = h_i(\underline{w}, t) + l_i(\underline{w}, t) \times u_i(\underline{w}, t) + d_i(t) \qquad ; \quad i = [1, 6] \quad (1)$$

Where $\underline{u}$ represents the vector of control inputs and $\underline{d}$ is the vector of external disturbances.
$\underline{h}$ and $\underline{l}$ are two vectors that represent the dynamics of the system, where their expressions are defined in table I.
The mass of the quadrotor is indicated by m, the moments of inertia are represented by $(I_x,\ I_y,\ I_z)$. The terms $(k_{ft_x},\ k_{ft_x},\ k_{ft_z})$ correspond to the translational drag coefficients, and $(k_{fa_x},\ k_{fa_y},\ k_{fa_z})$ represent the aerodynamic friction coefficients that influence the rotational dynamics.

TABLE I: Expressions of the dynamics of the system

Coefficient	Expression
h_1	$h_1 = -\frac{k_{ft_x}}{m}\dot{x}$
h_2	$h_2 = -\frac{k_{ft_y}}{m}\dot{y}$
h_3	$h_3 = -g - \frac{k_{ft_z}}{m}\dot{z}$
h_4	$h_4 = \frac{(I_y - I_z)}{I_x}\dot{\theta}\dot{\psi} - \frac{J_r\Omega}{I_x}\dot{\theta} - \frac{k_{fa_x}}{I_x}\dot{\phi}^2$
h_5	$h_5 = \frac{(I_z - I_x)}{I_y}\dot{\phi}\dot{\psi} - \frac{J_r\Omega}{I_y}\dot{\phi} - \frac{k_{fa_y}}{I_y}\dot{\theta}^2$
h_6	$h_6 = \frac{(I_x - I_y)}{I_z}\dot{\theta}\dot{\phi} - \frac{k_{fa_z}}{I_z}\dot{\psi}^2$
l_1	$l_1 = 1/m$
l_2	$l_2 = 1/m$
l_3	$l_3 = \frac{\cos\theta\cos\phi}{m}$
l_4	$l_4 = 1/I_x$
l_5	$l_5 = 1/I_y$
l_6	$l_6 = 1/I_z$

III. Proposed Control Strategy

This section will introduce the proposed control strategy for the quadrotor drone, whereby a combination of backstepping control and barrier Lyapunov stability will be emphasized. This control design incorporates the BOA approach to optimize the parameters of the controller [24], [28], [31]. The triangular observer is used to approximate the hidden states within the system [32].

A. Backstepping controller based on barrier Lyapunov

This part introduces the backstepping control method as the primary control strategy for the quadrotor drone. The backstepping method is implemented by defining a series of virtual control inputs for each state variable and gradually designing control laws that guarantee system stability at each stage of the process [24], [26].
In this study, the backstepping method is integrated with Barrier Lyapunov Functions (BLFs) to guarantee that the system's states remain within the predefined stability boundaries, enablingan accurate control performance, even in the presence of external disturbances and system uncertainties [28], [29].
Defining $\underline{w}_d$ as the vector of desired trajectories, we calculate

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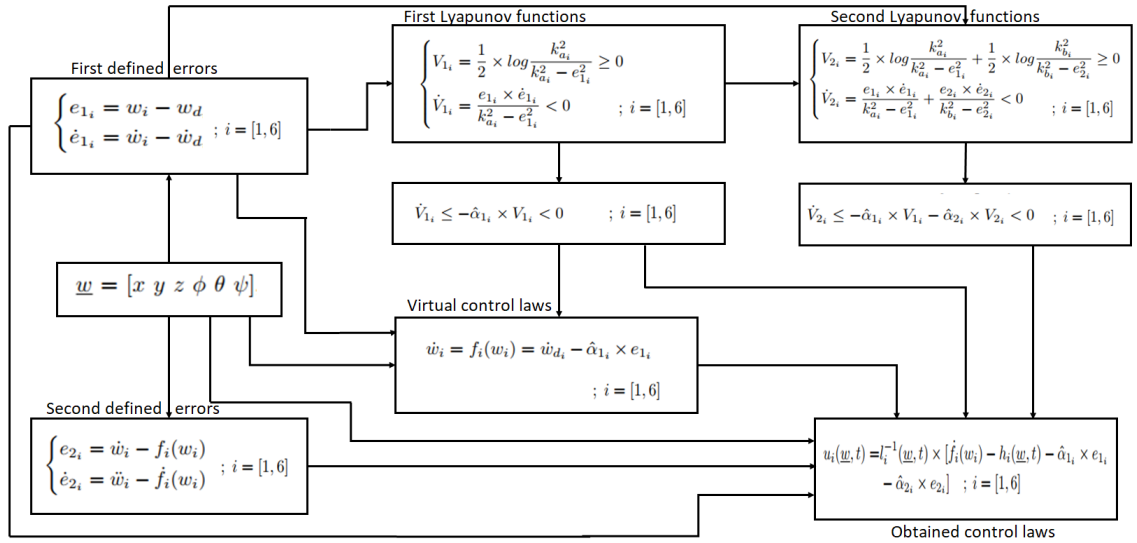

Fig. 2: Diagram of control law developement

the tracking errors as well as their derivatives in the following manner:

$$\begin{cases} e_{1_i} = w_i - w_{d_i} \\ \dot{e}_{1_i} = \dot{w}_i - \dot{w}_{d_i} \end{cases} \quad ; \ i = [1,6] \tag{2}$$

Subsequently, in order to analyse the system stability and to derive the virtual control laws, the barrier Lyapunov functions and their derivatives are defined in the following form :

$$\begin{cases} V_{1_i} = \dfrac{1}{2} \times log \dfrac{k_{a_i}^2}{k_{a_i}^2 - e_{1_i}^2} \geq 0 \\ \dot{V}_{1_i} = \dfrac{e_{1_i} \times \dot{e}_{1_i}}{k_{a_i}^2 - e_{1_i}^2} < 0 \end{cases} \quad ; \ i = [1,6] \tag{3}$$

Where k_{a_i}, $i = [1,6]$ is a positive coefficient. We assume that the virtual control laws can be written as follows :

$$\dot{w}_i = f_i(w_i) = \dot{w}_{d_i} - \hat{\alpha}_{1_i} \times e_{1_i} \quad ; \ i = [1,6] \tag{4}$$

Where $\hat{\alpha}_{1_i}$ is the optimal value of α_{1_i}, $i = [1,6]$ using BOA approach.
Then, according to (3) and (4), we obtain :

$$\dot{V}_{1_i} \leq -\hat{\alpha}_{1_i} \times V_{1_i} < 0 \quad ; \ i = [1,6] \tag{5}$$

The second step consists on defining new errors and their derivatives, determined in the following form :

$$\begin{cases} e_{2_i} = \dot{w}_i - f_i(w_i) \\ \dot{e}_{2_i} = \ddot{w}_i - \dot{f}_i(w_i) \end{cases} \quad ; \ i = [1,6] \tag{6}$$

Subsequently, an additional Lyapunov function is defined in order to evaluate the overall stability of the system and then to derive the control laws for each state.

$$\begin{cases} V_{2_i} = \dfrac{1}{2} \times log \dfrac{k_{a_i}^2}{k_{a_i}^2 - e_{1_i}^2} + \dfrac{1}{2} \times log \dfrac{k_{b_i}^2}{k_{b_i}^2 - e_{2_i}^2} \geq 0 \\ \dot{V}_{2_i} = \dfrac{e_{1_i} \times \dot{e}_{1_i}}{k_{a_i}^2 - e_{1_i}^2} + \dfrac{e_{2_i} \times \dot{e}_{2_i}}{k_{b_i}^2 - e_{2_i}^2} < 0 \end{cases} \quad ; \ i = [1,6] \tag{7}$$

From (7), the following inequality is obtained :

$$\dot{V}_{2_i} \leq -\hat{\alpha}_{1_i} \times V_{1_i} - \hat{\alpha}_{2_i} \times V_{2_i} < 0 \quad ; \ i = [1,6] \tag{8}$$

Where $\hat{\alpha}_{1_i}$ and $\hat{\alpha}_{2_i}$ are the optimal values of α_{1_i} and α_{2_i} respectively, $i = [1,6]$ using BOA approach. The control laws can subsequently be formulated in the following manner:

$$\begin{aligned} u_i(\underline{w},t) = & l_i^{-1}(\underline{w},t) \times [\dot{f}_i(w_i) - h_i(\underline{w},t) - \hat{\alpha}_{1_i} \times e_{1_i} \\ & - \hat{\alpha}_{2_i} \times e_{2_i}] \quad ; \ i = [1,6] \end{aligned} \tag{9}$$

Figure 2 illustrates the overall structure of control law derivation, summarizing the key equations and their interactions in the controller design.

B. Butterfly Optimization Algorithm

This part introduces the BOA as the optimization technique used to fine-tune the coefficients α_{1_i} and α_{2_i} of the backstepping controller. It is particularly effective in handling complex optimization problems due to its capacity to explore a vast search space and to maintain a balance between the exploration and exploitation phases [31].
The algorithm, based on two principal phases: global search and local search, represents the butterflies as individuals within a population, with their position in the search space corresponding to a potential solution to the optimization problem [31].
In this study, the BOA approach is employed to optimize two critical parameters in the backstepping control design: α_{1_i} and α_{2_i}, as they influence the stability and response speed of the system, and therefore affect the efficacy of the control performance. The use of BOA enables an efficient search for the optimal values $\hat{\alpha}_{1_i}$ and $\hat{\alpha}_{2_i}$, which minimize the tracking errors and enhance the overall stability of the quadrotor drone.

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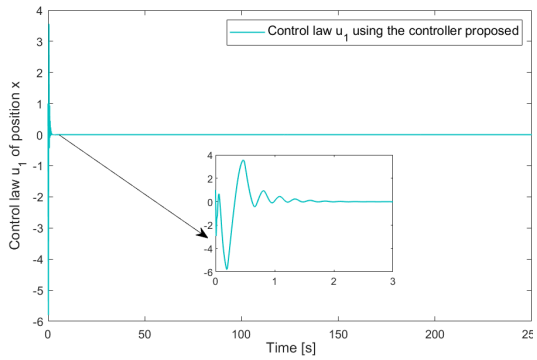
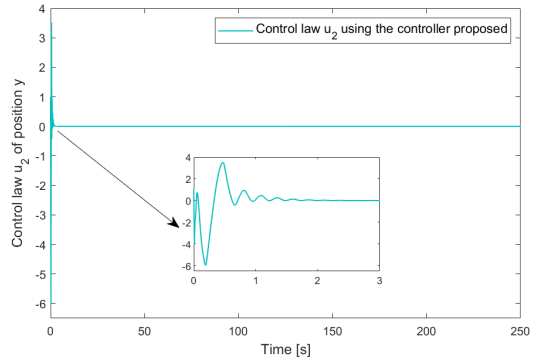
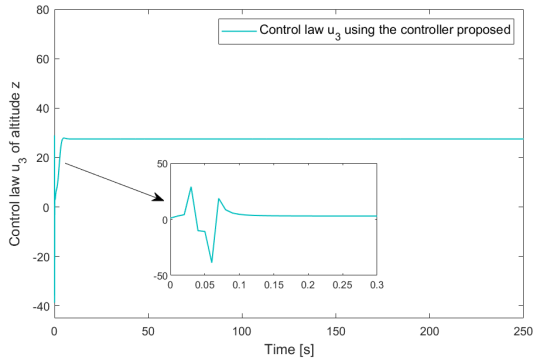


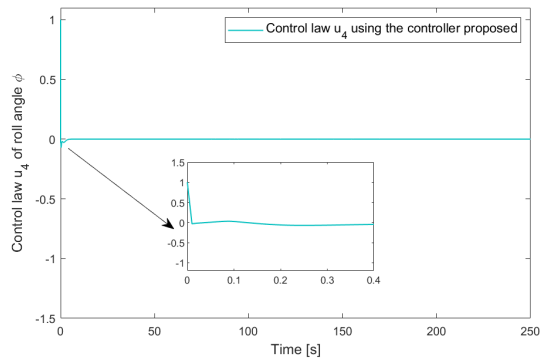
(a) The response of the control law u_1 achieved



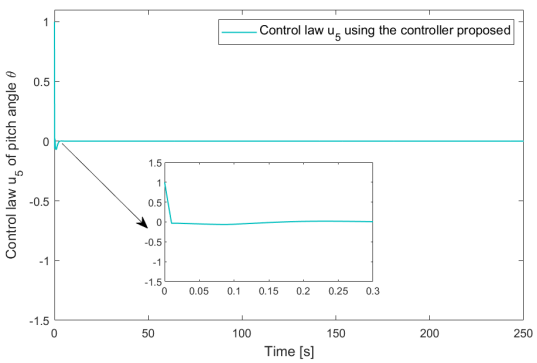
(b) The response of the control law u_2 achieved



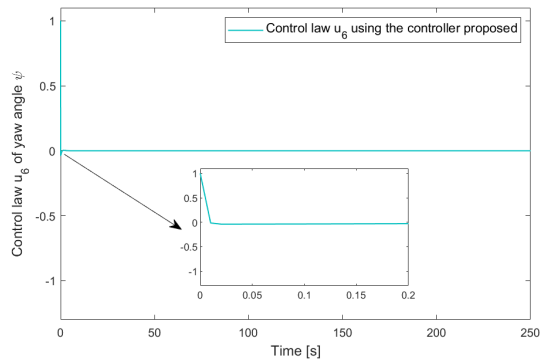
(c) The response of the control law u_3 achieved



(d) The response of the control law u_4 achieved



(e) The response of the control law u_5 achieved



(f) The response of the control law u_6 achieved

Fig. 3: The evolutions of the control laws achieved using the proposed controller

IV. Design of the Triangular Observer

The triangular observer, intended to estimate the hidden states of the quadrotor drone system, is presented in this section. This observer operates on the basis of the state feedback principle, whereby the known states of the system are employed to estimate the unknown states in a recursive manner [32], [39].

In this work, the triangular observer is employed to estimate the states that are critical for the accurate functioning of the control strategy, such as velocities. These estimated states are employed in the control loop to improve the overall performance of the controller [39].

Considering $\dot{\underline{\hat{w}}}$ as the vector of the approximated states using the triangular observer, the observer errors are then defined in the following form :

$$e_{o_i} = w_i - \hat{w}_i \qquad ; \quad i = [1, 6] \tag{10}$$

The approximated states $\dot{\underline{\hat{w}}}$ can be written as follows :

$$\begin{cases} \dot{\hat{w}}_i = \ddot{\hat{w}}_i + \lambda_{1_i} \times sign(w_i - \hat{w}_i) \\ \ddot{\hat{w}}_i = \hat{h}_i(\underline{\hat{w}}, t) + \hat{l}_i(\underline{\hat{w}}, t) \times u_i(\underline{\hat{w}}, t) + \lambda_{2_i} \times sign(\dot{w}_i - \dot{\hat{w}}_i) \\ \qquad ; \; i = [1, 6] \end{cases} \tag{11}$$

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Where $\underline{\lambda}_1$ and $\underline{\lambda}_2$ are two vectors of positive coefficients, and $\underline{\tilde{w}}$ represents the estimation vector, with :

$$\dot{\tilde{w}}_i = \dot{\hat{w}}_i + \lambda_{1_i} \times sign(w_i - \hat{w}_i) \qquad ; \; i = [1, 6] \tag{12}$$

V. Simulation results

This section presents a comparative analysis of the proposed control strategy, alongside two other methods: The first is the backstepping method using barrier Lyapunov without the BOA, the second method involves the traditional backstepping control based on the standard Lyapunov approach. The objective of the comparison is to evaluate the three methods in terms of their tracking accuracy, ability to reject disturbances, and overall system stability. Moreover, the analysis will focus on the states observed by the triangular observer, demonstrating the effectiveness of this approach in estimating the system's hidden states.

The first set of figures 3 illustrates the evolution of the control inputs u_1 to u_6 generated by the proposed controller. Each subfigure corresponds to one of the six control signals applied to control the system states: Specifically, u_1, u_2 and u_3 control the translational motion in the x, y and z axes, respectively, while u_4, u_5 and u_6 govern the rotational motion around the roll (ϕ), pitch (θ), and yaw (ψ) angles. These figures illustrate how the proposed controller dynamically adjusts the control inputs over time to ensure precise tracking of the desired state trajectories while maintaining system stability.

Figure 3 illustrates the efficacy of the proposed controller in minimizing control effort while guaranteeing system stability and precision across all six states.

Figure 4 illustrates the evolution of the system states - including the three translational positions (x, y, z) and the three rotational angles (ϕ, θ, ψ). Each subfigure presents both the desired trajectory and the actual trajectory achieved. The comparison highlights the tracking performance of the proposed control strategy, compared to the backstepping controller without BOA and the traditional Lyapunov-based backstepping approach.

The figures of trajectories 4 demonstrate the superior performance of the proposed controller using BOA approach, where it achieves the fastest response and convergence to the desired trajectories, followed by the method without BOA, which shows a slightly slower response. The traditional Lyapunov-based backstepping approach has the slowest convergence time, underscoring the effectiveness of the proposed strategy in improving system responsiveness.

Figure 5 presents the estimated states and observer errors for the six states of the quadrotor system: the three translational velocities ($\dot{x}$, $\dot{y}$, $\dot{z}$) and the three angular velocities ($\dot{\phi}$, $\dot{\theta}$, $\dot{\psi}$). Each subfigure illustrates three curves: the hidden state, the estimated state obtained by the triangular observer and the corresponding observer error $\dot{e}_{o_i}$, which represents the difference between the actual and estimated values.

The efficacy of the triangular observer is clearly demonstrated by its capacity to accurately estimate the hidden states of the quadrotor system. The observer exhibits robust performance by minimizing estimation errors.

VI. Conclusion

This paper proposes a control strategy combining backstepping control with barrier Lyapunov stability for the purpose of controlling a quadrotor drone. The objective of this method was to ensure precise trajectory tracking while enhancing the system's robustness against external disturbances and mitigating uncertainties. The Butterfly Optimization Algorithm (BOA) was employed in order to optimize the controller's key parameters of the backstepping controller. Furthermore, a triangular observer was introduced to approximate the hidden states of the system. The simulation results demonstrated the efficacy of the proposed control strategy, exhibiting considerable enhancements in trajectory tracking precision and robust behavior in the presence of external disturbances. Nevertheless, although the results are promising, there are still some limitations to this work as the computational complexity of the BOA approach, particularly in real-time applications, may present a challenge for computing systems.

Further research could be directed towards the refinement of the optimization methodology in order to reduce the computational requirements, and the application of the method to experimental systems for real-world validation.

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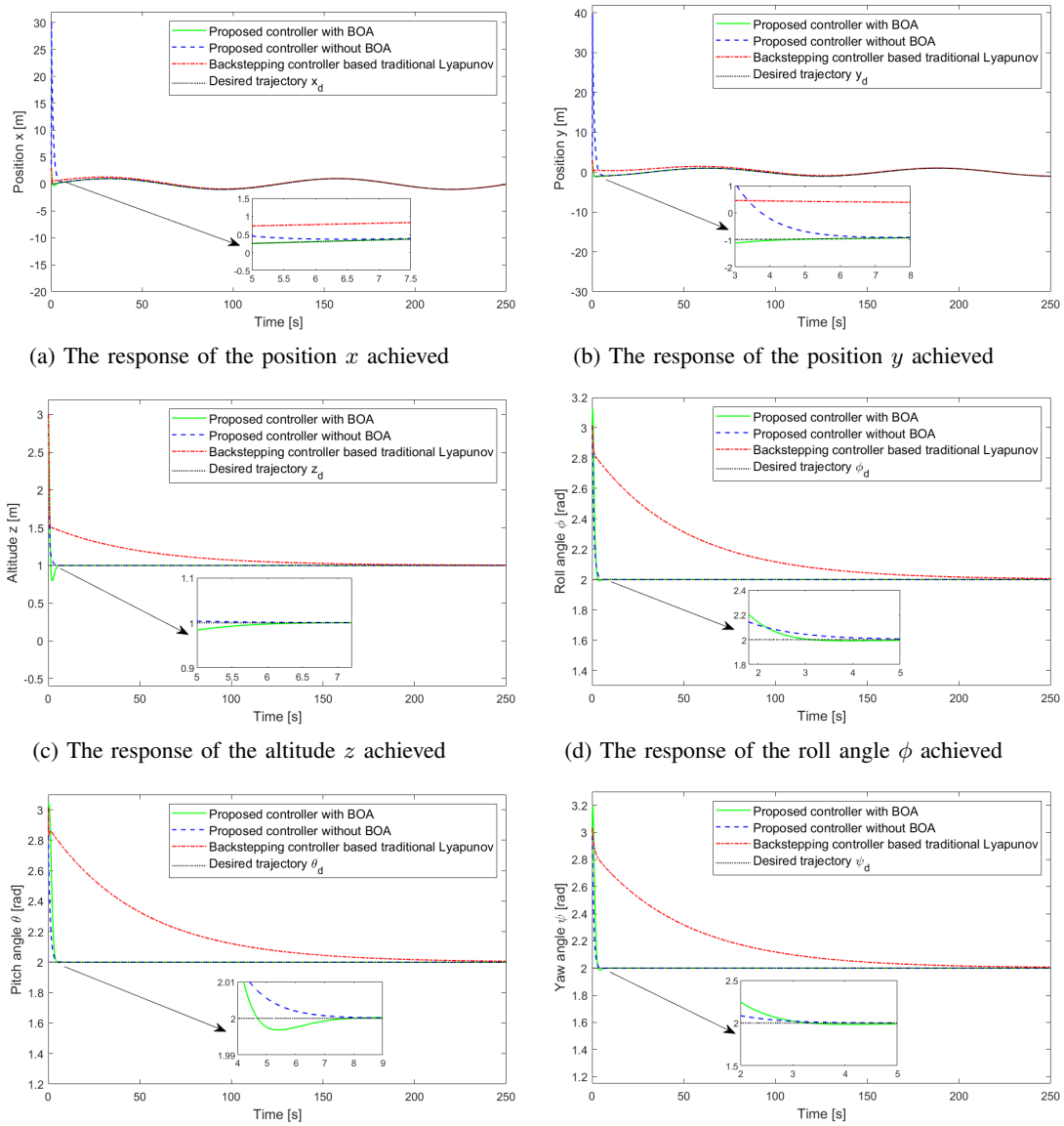

(a) The response of the position x achieved

(b) The response of the position y achieved

(c) The response of the altitude z achieved

(d) The response of the roll angle ϕ achieved

(e) The response of the pitch angle θ achieved

(f) The response of the yaw angle ψ achieved

Fig. 4: The evolutions of the trajectories achieved and the desired trajectories using the three methods

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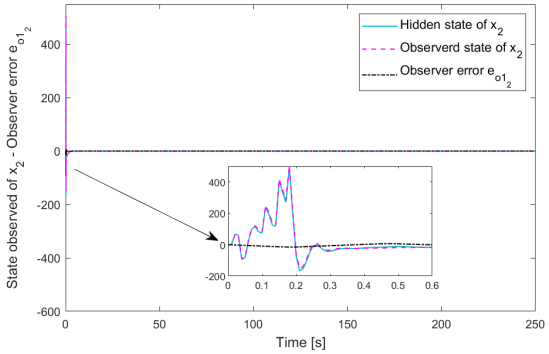
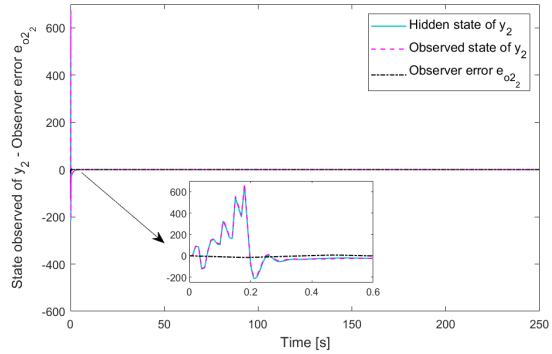
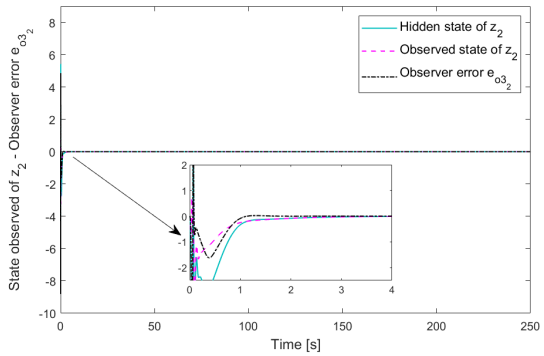


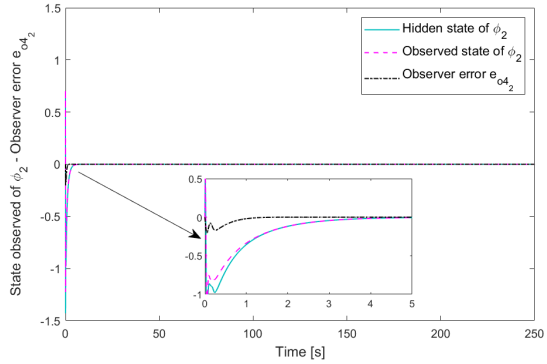
(a) The response of the observed state $\dot{\hat{x}}$ and the observer error $\dot{e}_{o_1}$ achieved



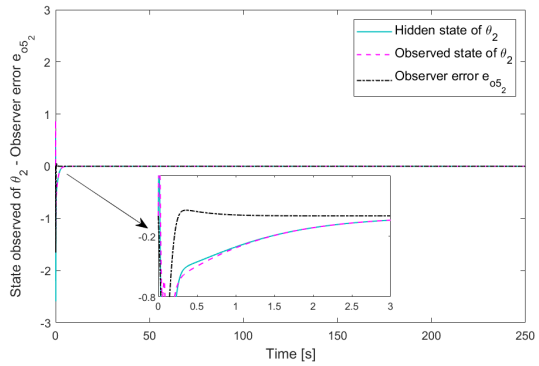
(b) The response of the observed state $\dot{\hat{y}}$ and the observer error $\dot{e}_{o_2}$ achieved



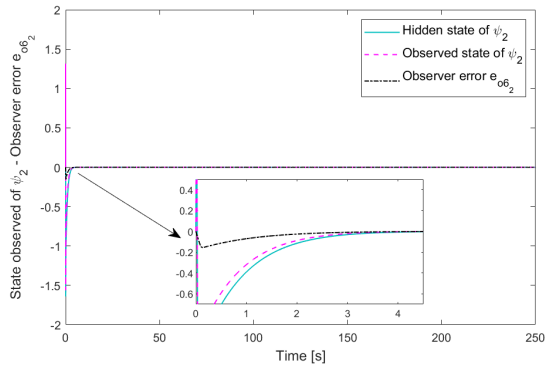
(c) The response of the observed state $\dot{\hat{z}}$ and the observer error $\dot{e}_{o_3}$ achieved



(d) The response of the observed state $\dot{\hat{\phi}}$ and the observer error $\dot{e}_{o_4}$ achieved



(e) The response of the observed state $\dot{\hat{\theta}}$ and the observer error $\dot{e}_{o_5}$ achieved



(f) The response of the observed state $\dot{\hat{\psi}}$ and the observer error $\dot{e}_{o_6}$ achieved

Fig. 5: The evolutions of the hidden states, the estimated states and the observer errors using the triangular observer

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